



Mechanical Technology

Introduction to welding

Text Book Chapter 1

Page

Introduction:

The Welding concept and the work place layout

What is welding



- ▶ Welding is a fabrication process whereby two or more parts are fused together by means of heat, pressure or both forming a join as the parts cool.
- ▶ Welding is usually used on metals and thermoplastics

Improving safety through layout and design

- ▶ Where possible, welding and grinding tasks should be done in the best working zone (between the shoulders and knees).
- ▶ Employers should ensure tools used by workers are:-
 - Maintained in good working condition
 - Suitable for the tasks undertaken
 - Suitable for the work area where the task is to be performed
 - Elevated welders and leads eliminate the need to use high force when moving welders and greatly reduces the risk of tripping



Improving safety through layout and design in the workshop 1

- Use welding booths to prevent surrounding workers from being exposed to welding flash arc or use curtains and screens.
- Pipe welding gases into the building to eliminate the need for workers to manually handle cylinders when they need replacing.
- Where cylinder gases are used, ensure they are stored in a safe location and secured to prevent tip-over.
- Ensure welding work area is appropriately ventilated through natural airflow or a fixed/portable extraction system.
- Welding masks are located close to where welding tasks are performed



Improving safety through layout and design in the workshop 2

- **Observe the safety procedure with entering and exiting a workshop.**
- When exiting the workshop Check that any tools you have been using have been put away in the appropriate spots, clean up your work area and notify the workshop staff.

Identify the safety signs that indicate the doors used for evacuation

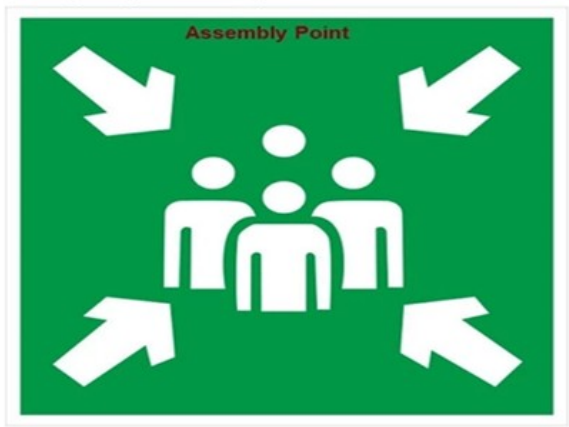
- Emergency sign always have a green background pasted on doors, exits, escape routes, equipment and facilities, etc



Welding Workshop

- ▶ Escape routes and doors must be labelled clearly with a sign or notice reading 'no exit', 'no way out', exit or labelled with their use, such as 'storeroom'

Emergency Assembly Point



Emergency signs



Safety evacuation drill and perform it

Learners practicing a fire drill in a smoke trailer with a firefighter assisting them. This is aimed at teaching the children proper procedures for evacuating the building in a real life fire or emergency situation.



FIRE DRILLS AND FIRES

1. All learners, teachers, and other employees shall be required to leave the school building in an orderly and rapid manner.

➤ Teachers shall ascertain that no learner remains in the building.

2. Teachers shall be prepared to select alternate exits and direct their classes to these exits in the event the designated escape route is blocked.

In case of an actual fire emergency, the following actions will be taken:

1. Sound fire signals.

2. Call fire departments and police.

3. Learners and adults evacuate the building to outside assembly areas.

4. In outside assembly areas, teachers shall take roll, report missing students, and provide assistance to injured learners.

5. If the fire is serious, learners shall be taken to an alternate location for protective custody until parents/guardians can pick them up or until they can be safely transported to their homes.

TYPES OF EVACUATION

- A total building evacuation also named single-stage
- Evacuation from the dangerous area to a place where people can follow an escape route;
- Staff alarm which is an evacuation directed by staff (usually in shops or cinemas)
- Warning / alert signal evacuate. This type will allow for some time to investigate the potential source of fire followed by an evacuation signal if appropriate

Week Two

Safety, First Aid HIV & AIDs

(OHS Act)

Basic Safety

Workshops is a dangerous place. This is because there are machines and equipment that are designed to cut, join, and shape hard materials and metals.

To ensure safety, you want to be in an environment where you are free from injuries, the risk of being injured, or injuring your friend or a colleague



SAFETY PROCEDURES

Workplace safety procedures are standardized processes that outline how to conduct tasks with minimal risk to people, materials, and work environments. The procedures include work-related tasks that address safety concerns, safety equipment, and work-area precautions.

Safety can be improved if we can all ensure

That:-

- ☐ The workshop remains clean and neat.
- ☐ Equipment and materials are stored in their proper places.
- ☐ The first aid kit is easily accessible and contains the necessary medical items.
- ☐ Fire extinguishers are maintained and in good working order.

What is an accident?



- An accident is an unplanned and uncontrolled event that happens because of unsafe acts and/or unsafe conditions causing loss.
- Accidents have consequences. It can hurt you. It can hurt somebody else. It can cost money. It can break equipment. It can cause people to lose their jobs.
- Accidents are BAD, and we want to avoid it all costs. That is why we always keep SAFETY FIRST!
- Accidents happen due to unsafe conditions or unsafe actions or both

What is an unsafe act?

An **unsafe act** is a human error or personal mistake by which there is a chance of accident, injury, property loss, or harm to the environment

An unsafe act is a human activity resulting in accident injury, etc. The **unsafe act** might be deliberate or committed due to the absence of knowledge, ignorance, or forgetfulness.



Unsafe acts can include:

1. Operating without qualification or authorization.
2. Operating equipment at unsafe speed.
3. Failure to warn.
4. Bypass or removal of safety devices.
5. Using defective equipment.
6. Use of tools for other than their intended purpose.
7. Working in hazardous locations without adequate protection or warning.
8. Improper repair of equipment.
9. Horseplay.
10. Wearing unsafe clothing.
11. Taking an unsafe position.

Unsafe Condition :-

Unsafe condition are hazards that have the potential to cause injury or death to an employee. Some of these hazards include erroneous safety procedures.

- Defective tools, & equipment
- Inadequate supports or guards.
- Congestion in the workplace.
- Inadequate warning systems.
- Fire and explosion hazards.
- Poor housekeeping.
- Hazardous atmospheric condition.
- Excessive noise.
- Poor ventilation.
- Rough and slippery floors
- Insufficient light in a workshop
- Badly planned workshop



Good housekeeping

- ❑ Good housekeeping in a workshop simply means an orderly arrangement of tools, equipment, operations, storage facilities and materials
- ❑ To put it in simpler words, housekeeping can be defined as everything in its place and a place for everything.



Basic First Aid



First aid is emergency care given immediately to an injured person. The purpose of first aid is to minimize injury and future disability. In serious cases, first aid may be necessary to keep the victim alive.

The decision to act appropriately with first aid can mean the difference between life and death. Begin by introducing yourself to the injured or ill person.

The first step in any emergency is the recognition of the problem and providing help. You must know where the first aid kit is. Ask another learner to get the first aid kit, while you are rendering assistance.

Protective Clothes



How to dress Small Cuts

- Clean the wound with running water.
 - Wash the skin around the injury with soap.
 - Don't worry if soap gets into the wound, though it is likely to sting and irritate the raw tissue.
 - Rinse the wound thoroughly to rid it of any dirt and soap.
 - Use tweezers to remove particles (like broken glass or gravel)
 - Dry off the area, wiping from the wound outwards.
 - Apply antiseptic directly to the wound
 - Cover with a plaster.
- 

How to dress Minor Burns

A minor burn is damage to the outer and underlying layer of skin from thermal, chemical, electrical, or radiation exposure. They can cause pain, swelling, blisters. Cool the burn.

TREATING MINOR BURNS

1

Remove all clothing and jewellery that **isn't stuck to the burn** from around the burn area.



2

Place the burn under cool, **running water** for 20 minutes.

Do not apply ice or ice packs.



3

Cover the area with a sterile, **dry dressing** if the burn is oozing, unless on the face.



4

Seek medical attention if your burn is bigger than a 50 cent piece.



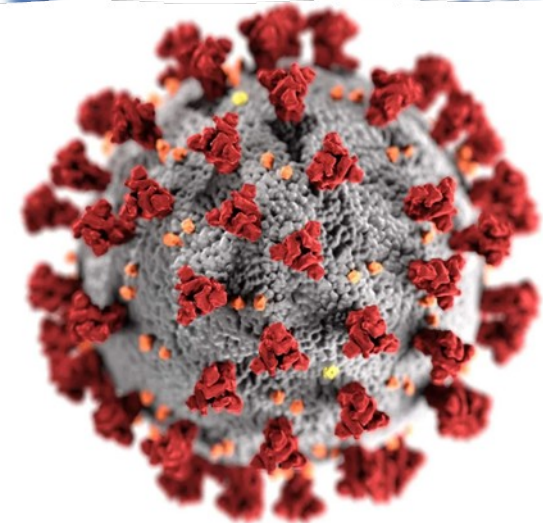
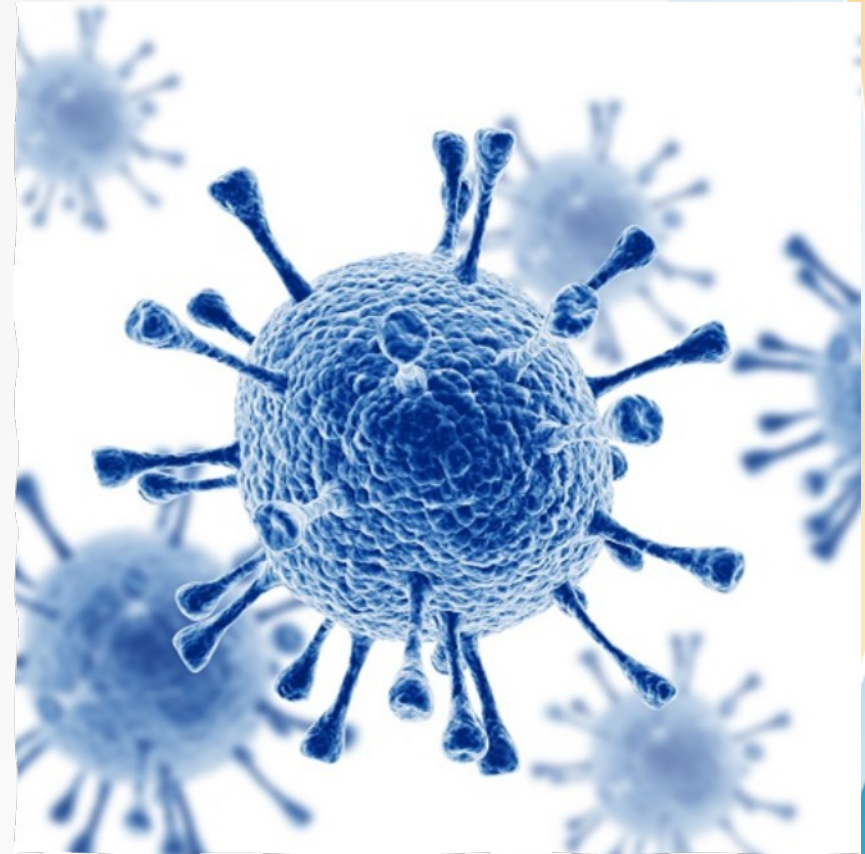
Disease

HIV Aids

Human immunodeficiency virus infection / acquired immunodeficiency syndrome (HIV/AIDS) is a disease of the human immune caused by infection with human immunodeficiency virus (HIV).

Covid

Most people infected with the virus will experience mild to moderate respiratory illness and recover without requiring special treatment.

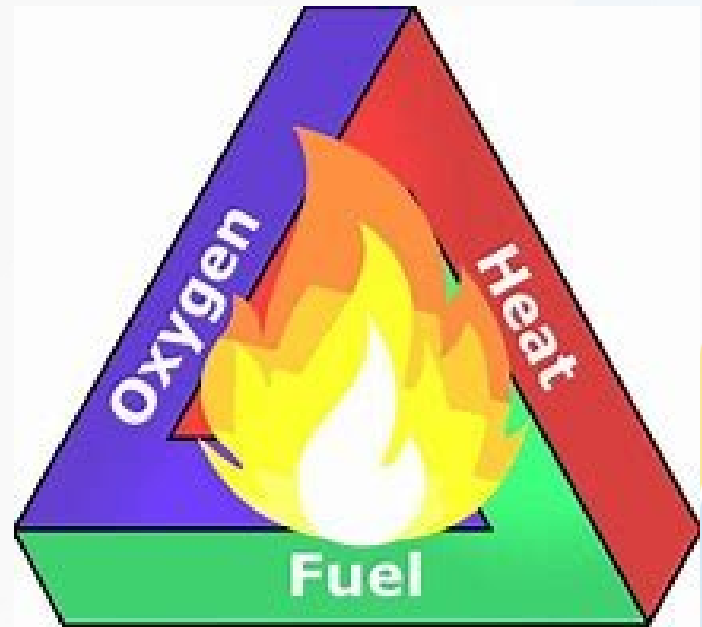


Causes of Fires

FAULTY ELECTRICS AND EQUIPMENT
WASTE AND COMBUSTIBLE MATERIAL
SMOKING

Arson is the deliberate act of starting a fire

HUMAN ERROR



Fire Safety Precautions

- Be Prepared
- Identify Fire Hazards
- Do something about it



Fire Extinguishers

Class A:

Wood, paper, etc.

Class B:

Flammable liquids such as petrol, oil and paraffin.

Class C:

Electrical fires.

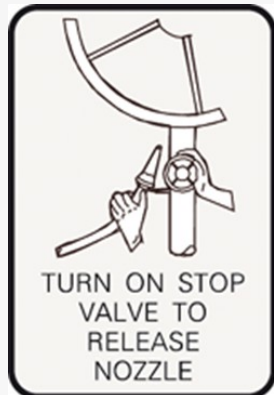
Class D:

Flammable metals such as magnesium, lithium, etc

KNOW YOUR FIRE EXTINGUISHER				
CHOOSING THE RIGHT EXTINGUISHER CAN PREVENT PROPERTY DAMAGE AND SAVE LIVES				
Extinguisher Type → Type of Fire ↓	 Water	 Foam	 CO ₂	 Dry Chemical
A Paper, Wood & Plastic 	✓	✓	✗	✓
B Flammable & Combustible Liquids 	✗	✓	✓	✓
C Electrical Equipment 	✗	✗	✓	✓

Fires can be extinguished :-

- By cooling water is used to cool the burning material below the temperature at which it starts to burn.
- By smothering carbon dioxide (CO₂) or foaming agents are used to smother the burning material so that air is excluded.
- By removing the fuel. This is usually very difficult to do. An example is turning off a fuel line.
- By disrupting the chemical chain reaction or interrupting the flame.



Dangerous Items

Gas Bottles

- All kinds of gas are used in workshops. From inert gas, which may not be flammable, but it may be dangerous such as CO₂, although you cannot see or smell it, it can kill you.

Power Tools

- Power tools should be stored in a designated area. The designated area should be demarcated per power tool.
- Some power tools come supplied in a handy carry case which makes it easier to store.

Other Hazardous Materials

- Hazardous substances include paints, solvents, lead, powders, lacquers, paint removers, resins, battery acid, used oils, adhesives, degreasers, surface preparation products, rust converters and removers, and dusts.

Questions & Answers

